

		When v is doubled, λ is halved.
30.	D	Background = 5 s^{-1} ; Actual Count rate $C = 39 - 5 = 34 \text{ s}^{-1}$ 20 minutes later (2 half-lives) Actual $C = 34 (0.5)^2 = 8.5 \text{ s}^{-1}$ $C \propto 1/r^2$ $\frac{C_2}{C_1} = \frac{r_1^2}{r_2^2} = \left(\frac{40}{20}\right)^2 = 4$ Reading on detector $C_2 = 4 \times 8.5 + 5 = 39.0 \text{ s}^{-1}$